StructOptima

Formulas, decisions and edge cases — as implemented

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Preliminary design aid. Every value below is what the code computes. Code values were compared with the printed standards on 20 October 2026 (evidence: `docs/verification/standards-check-2026-10-20.md`); that comparison is not an engineer's verification of any design. Every output must be checked by a licensed structural engineer before construction.

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0 Revision history

Newest first. Ids point to the records in `docs/assurance/` (N = finding, D = decision, RM = remediation).

Date	Change
21 Oct 2026 (later)	**DXF corpus** (74 named drawings + seeded random buildings) found 8 defects, fixed (N53–N60): mirrored blocks read mirrored, R12 polylines crashed, unitless / R12 / survey drawings at the wrong scale, circular columns on layer 0 dropped, rotated plans misread, 1–2 columns crashed the dynamic analysis, repeated warnings, outputs broken by a column beyond capacity. Decisions taken under the owner's delegation: D-33 (scope: a slab panel is required), D-34 (rotated plans), D-35 (unit inference), D-36 (circular columns), D-37 (column orientation deferred), D-38 ("beyond capacity" display); Section 11, Section 12. No formula changed. Tests 1154 passed, 3 skipped, 3 xfailed.
21 Oct 2026	**Geometry bound (D-32, approved by the owner):** a column / beam / wall coordinate that is not finite, or a structure wider than 1000 m (raw drawing: more than 10^6 units), is an input error before any calculation (Section 11). Found by the DXF fuzz tests: a line 10^{11} units long made the automatic column placement create 3×10^8 nodes (OOM, N50) or a design with $V_B \approx 8 \times 10^{14}$ kN. Non-finite design forces now stop the design instead of passing a column (N51) or reading as an input error (N52); Section 12. No formula changed. Tests 1029 passed, 3 skipped, 3 xfailed.
20 Oct 2026 (evening)	D-28 and the N46 fix approved by the owner. **N46 fixed:** the half beam width of each adjacent thicker panel is taken off the beam web (beam loads, column takedown, seismic weight; Section 4). **D-29 (owner request):** columns in 230 mm wall lines get a 230 mm side, within the code limits (Section 7.3). New check: beam wider than the column it frames into. Appendix A values updated (W, V_B, B5, C5, footing). Tests 809 passed.
20 Oct 2026 (later)	Appendix A: worked hand check of `complex_house.dxf` (locked by `test_hand_check_complex_house.py`); all values agree with the printed formulas. Findings: beams are designed for their worst floor (now stated in Section 8); 0.5 kN/m of slab over beams counted twice (N46, conservative, fix PROPOSED); drawn column sizes and room labels outside the building were not reported. **DXF reading (D-28, proposed):** layer roles from names and drawn shapes, editable on the Setup page with a plan preview; plinth / lintel layers ignored; drawn-vs-designed WARN; labels outside the structure reported (Section 11). Tests 798 passed; 92-drawing sweep unchanged.
20 Oct 2026	D-27 approved by the owner (soft-storey tolerance 10 %). All documentation updated for the dynamic analysis. This document created.
20 Oct 2026	**Dynamic analysis** (RM-36, D-26): response spectrum method (IS 1893 7.7.5) on a rigid-diaphragm 3D model with design eccentricity (7.8.2), torsional irregularity (Table 5(i) Amd 2), irregular modes (Table 6(vii) Amd 2), soft storey with the 0.2 % drift limit; "Seismic method" FAIL replaced by WARNs naming what the model cannot do. CI installs `pypdfium2`.
20 Oct 2026	**Standards check** (RM-35, N29–N45) against the printed IS 456, IS 1893 (Part 1):2016 (Amd 1, 2), IS 13920:2016 (Amd 1, 2), IS 875 (Parts 1–3), IS 1786, NBC 2016 Part 4 and SP 16: static spectrum (S1), minimum V_B (S3), 2016 method rule (S4), OMRF in Zones III–V (S2), soft storey / mass irregularity / floating columns (S5–S7), IS 13920 joint factor and breadth, roof SCWB, grade, steel, column geometry and axial stress, hooks, spacing (J1–J9), Table 26 case 4 (T1), Table 19 M15 (T2), exposure and fire (D1, F1, F2), office load (L1), wind k_2 and $h/w \geq 6$ (W1, W2), rigid-diaphragm load sharing (A1), DXF import (X1–X4), audit placeholder (P1), quick reference (Q1). Decisions D-21, D-22, D-24, D-25 approved.
20 Oct 2026	**UI restructure** (RM-34, D-20): seven pages, inputs kept across pages, out-of-date results flagged; display defects N23–N26 (footing punching always "OK", header status, slab status, stale results).
16 Oct 2026	Storey drift (N16), no simulated outputs (N14, N15), status semantics (N17), confirmed site inputs (N18), drawing set, IFC export, 3D viewer, repository clean-up (D-07), automatic column growth for drift (D-16).
15 Oct 2026	Audit and remediation of findings R1–R27, N1–N13: single design pipeline, combinations on 2D frames, P–M column design, status aggregation.

1 Conventions

- Units: kN, m, s for loads, geometry and analysis; mm for section dimensions; N/mm^2 for stresses.
- $E_c = 5000\sqrt{f_{ck}}$ N/mm^2 (IS 456 6.2.3.1); $E_s = 200$ kN/mm^2 (5.6.3); $g = 9.81$ m/s^2.
- Loads on columns (`load_kn`) are *service* loads; analysis envelopes and design forces are *factored*.
- Status of every check: **FAIL** (not acceptable as is), **WARN** (needs an engineer's review), **PASS** (only the implemented checks passed — never "verified"). The building status is the worst item.

2 Inputs, defaults and decisions

2.1 Inputs (Setup page)

Input	Default	Notes
Geometry	manual 18 m × 12 m, 2 storeys × 3.5 m	or DXF / IFC (Section 11)
Occupancy	Residential	imposed loads, Table 4.1
SBC	**none — required**	from the geotechnical report; sizes every footing
Seismic zone / soil	III / II	site confirmation is cleared when any site parameter changes
Structural system	OMRF ($R = 3$)	OMRF is a FAIL in Zones III–V (IS 1893 Table 9 Note 1)
Importance factor I	by occupancy	Table 8; 1.2 for > 200 occupants is the engineer's input
Wind zone / terrain	2 / 2	IS 875-3
Concrete / steel	M25 / Fe415	
Exposure condition	Moderate (D-21)	IS 456 Table 3 ⇒ covers, minimum grade
NBC type of construction	Type 2 (D-22)	NBC 2016 Part 4 Table 1 ⇒ fire ratings
Cracked sections	on	off gives a WARN (IS 1893 6.4.3.1)
Fixed bases	on	pinned bases often create a soft ground storey

2.2 Decisions

Id	Decision	Status
D-02	OMRF default; SMRF on user selection with warning	superseded in Zones III–V by Table 9 Note 1
D-03	Importance factor defaults by occupancy	proposed (checked against Table 8)
D-04	Effective length $l_e = 1.0\,l$ (Annex E not implemented)	open — engineer to decide
D-07	Repository clean-up	approved 16 Oct
D-16	Automatic column growth when EQ drift fails	approved 16 Oct
D-20	UI on Streamlit with seven pages	approved 20 Oct
D-21	Exposure input, default Moderate; covers = max(Table 16, Table 16A)	approved 20 Oct
D-22	NBC type of construction input, default Type 2	approved 20 Oct
D-24	Office imposed load 4.0 kN/m^2 (no separate storage)	approved 20 Oct
D-25	Lateral sharing by condensed frame stiffness with slab links	approved 20 Oct
D-26	Response spectrum on a rigid-diaphragm 3D model for every building; what the model cannot represent is a WARN with the external check named	approved 20 Oct
D-27	Soft storey when computed stiffness is > 10 % below the storey above	approved 20 Oct

Id	Decision	Status
D-28	DXF layer roles from the layer name and the drawn shapes; engineer confirms on Setup; plinth / lintel layers ignored by default; drawn-vs-designed columns and labels outside the structure reported	approved 20 Oct
D-29	Residential columns in 230 mm wall lines: 230 mm side across the wall, within IS 13920 7.1.1 (Zone II OMRF only), IS 456 Fig. 1 (50 % exposed) and 32.1 (long side ≤ 900)	requested by the owner 20 Oct

3 Geometry

- **Bays (manual input):** $n = \max(1, \lceil L/5 \rceil)$; if $L/n < 3$ reduce n while $L/(n-1) \leq 5$; grid lines at iL/n (project practice, not a code rule).
- **Tributary area:** $A = (\frac{1}{2}l_{\text{left}} + \frac{1}{2}l_{\text{right}})(\frac{1}{2}l_{\text{below}} + \frac{1}{2}l_{\text{above}})$; staircase void corners lose $\frac{1}{4}$ bay and get 15 kN/m^2 $\times$ lost area (80 % DL, 20 % LL).
- **Slab panels:** a bay is a panel if it is not a void and has columns at its four corners or its centre is enclosed by walls/beams on four sides; discontinuous edges from neighbouring panels.

4 Loads

4.1 Occupancy (IS 875 Parts 1 and 2)

Occupancy	LL	Roof	Corridor	Finish	Partition	Services	t (mm)
Residential	2.0	1.5	3.0	1.0	1.5	0.5	125
Commercial (Office)	4.0	1.5	4.0	1.0	1.0	1.0	150
Commercial (Retail)	4.0	1.5	5.0	1.5	0.5	1.0	150
Institutional	3.0	1.5	4.0	1.0	1.0	1.5	150
Assembly Hall	5.0	1.5	5.0	1.5	0.5	2.0	175

kN/m^2. Office 4.0 = IS 875-2 Table 1 v)(b) "without separate storage" (2.5 with storage) — D-24. The corridor value is used for the staircase only; corridors and balconies on the floors carry the floor load (gap — IS 875-2 gives them higher values).

$$q_{DL,\text{floor}} = 25t + \text{finish} + \text{partition} + \text{services}, \qquad q_{DL,\text{roof}} = 25t + \text{finish} + \text{services}$$

Panels designed thicker than the occupancy slab add $25(t_{\text{designed}} - t)$.

Walls: $w = t\,h\,(1 - \text{openings})\,\gamma$, $\gamma = 19.2$ (brick; IS 875-1: 18.85, conservative), 7.5 (AAC), 25 (RCC) kN/m^3; $h = h_{\text{storey}} - 0.125 - 0.400$ m; openings 0.33 default.

Column takedown (service, cumulative from the roof): $DL = Aq_{DL}+$ panel extras + cantilever + walls ($\frac{1}{2}$ to each end column; parapet 0.33$\times$ storey wall) $+\frac{1}{2}$ beam webs $b(D-t)25L$ + column self-weight $bDh \cdot 25$ + stair; $LL = Aq_{LL} + \ldots$ Ground-storey walls go to the footing (plinth).

Beam UDL from slabs: two-way ($l_y/l_x \leq 2$): short-edge beam $k = l_x/3$, long-edge beam $k = \frac{l_x}{6}\left(3 - (l_x/l_y)^2\right)$; one-way: $l_x/2$ on long edges; $w = kq+$ walls + web, with

$$\text{web} = b(D-t)\,25 - \sum_{\text{adjacent panels}} \frac{b}{2}\,25\,(t_{\text{panel}} - t) \qquad \text{(N46, fixed 20 Oct 2026)}$$

because the extra thickness of a panel designed thicker than the occupancy slab t is already counted up to the beam centre line. The same web is used in the column takedown and the seismic weight.

4.2 Seismic weight (IS 1893 7.3, 7.4, Table 10)

$$W_j = Aq_{DL} + \text{webs} + \tfrac{1}{2}(\text{walls} + \text{columns})_{\text{below}} + \tfrac{1}{2}(\ldots)_{\text{above}} + A\,q_{LL}\,f, \qquad f = \begin{cases} 0.25 & q_{LL} \leq 3 \\ 0.50 & q_{LL} > 3 \\ 0 & \text{roof} \end{cases}$$

4.3 Combinations (IS 456 Table 18, IS 1893 6.3.2.1)

$1.5(DL+LL)$; $1.2(DL+LL\pm WL)$; $1.5(DL\pm WL)$; $0.9DL\pm1.5WL$; $1.2(DL+LL\pm EL)$; $1.5(DL\pm EL)$; $0.9DL\pm1.5EL$, each direction separately (orthogonal frames). Not used: $100+30\,\%$ (6.3.2.2, non-orthogonal systems), vertical EQ (6.3.3.1).

5 Structural analysis

5.1 Plane frames

Frames on every grid line (XZ at constant y, YZ at constant x); 6×6 plane frame elements (EA/L, $12EI/L^3$, $6EI/L^2$, $4EI/L$, $2EI/L$). Cracked stiffness (IS 1893 6.4.3.1): $0.70I_g$ columns, $0.35I_g$ beams. Nodes of a floor not connected by a beam are tied by axially rigid links ($A=10$ m^2, $I\approx0$) representing the slab in its plane. Mechanism: free DOF with $K_{ii}<10^{-9}\Rightarrow$ UNSTABLE; cond $>10^{12}\Rightarrow$ SINGULAR; beams then use $wL^2/8$.

5.2 Condensed lateral stiffness and the 3D model

Each frame is solved for a unit horizontal load at each floor: flexibility $\mathbf{F}_f$, stiffness $\mathbf{K}_f=\mathbf{F}_f^{-1}$ ($N\times N$). Floor DOFs (u_x,u_y,θ) about the origin:

$$u_{f,i}=u_{x,i}-y_f\theta_i\ \text{(XZ)},\qquad u_{f,i}=u_{y,i}+x_f\theta_i\ \text{(YZ)},\qquad \mathbf{K}_3=\sum_f \mathbf{T}_f^{\mathsf T}\mathbf{K}_f\mathbf{T}_f.$$

Frame forces under floor loads $\mathbf{F}$: $\mathbf{f}_f=\mathbf{K}_f\mathbf{T}_f\mathbf{K}_3^{-1}\mathbf{F}$. If $\mathbf{K}_3$ is not positive definite (fewer than two frame lines in a direction) the storey forces are shared in translation only ($\sum\mathbf{K}_f\,\mathbf{u}=\mathbf{F}$) and a WARN is given.

Mass: per floor $m=W/g$ at the slab-panel centroid (x_{cm},y_{cm}), rotational inertia mr^2,

$$r^2=\frac{\sum A_p\left[(x_p-x_{cm})^2+(y_p-y_{cm})^2+(a_p^2+b_p^2)/12\right]}{\sum A_p}.$$

5.3 Static seismic forces (IS 1893 6.4.2(a), 7.2, 7.6)

$$A_h=\frac{Z}{2}\,\frac{I}{R}\,\frac{S_a}{g},\qquad \frac{S_a}{g}=\begin{cases}2.5 & 0<T\le T_1\\ N/T & T_1<T\le4\\ c & T>4\end{cases}\qquad (T_1,N,c)=\begin{cases}(0.40,1.00,0.25) & \text{soil I}\\ (0.55,1.36,0.34) & \text{II}\\ (0.67,1.67,0.42) & \text{III}\end{cases}$$

$Z=0.10,0.16,0.24,0.36$ (Zones II–V); $R=3$ (OMRF), 5 (SMRF).

$$T_a=0.075h^{0.75}\ \text{(bare)};\quad T_a=\min\!\left(0.075h^{0.75},\ 0.09h/\sqrt{d}\right)\ \text{(infill, 7.6.2.1)},\ d=\text{larger plan dimension}$$

$$V_B=\max(A_h,\rho)\,W,\quad \rho=0.7,1.1,1.6,2.4\,\%\ \text{(Table 7)},\qquad Q_i=V_B\frac{W_ih_i^2}{\sum W_jh_j^2}\ \text{(7.6.3)}.$$

5.4 Response spectrum method (IS 1893 7.7.5) — used for every building (D-26)

$$\mathbf{K}_3\boldsymbol{\phi}=\omega^2\mathbf{M}\boldsymbol{\phi},\qquad \Gamma_k=\frac{\boldsymbol{\phi}_k^{\mathsf T}\mathbf{M}\mathbf{r}}{\boldsymbol{\phi}_k^{\mathsf T}\mathbf{M}\boldsymbol{\phi}_k},\qquad M_{\text{eff},k}=\frac{(\boldsymbol{\phi}_k^{\mathsf T}\mathbf{M}\mathbf{r})^2}{\boldsymbol{\phi}_k^{\mathsf T}\mathbf{M}\boldsymbol{\phi}_k}$$

$$A_k=\frac{Z}{2}\,\frac{I}{R}\,\frac{S_a}{g}(T_k)\ \text{with}\ \frac{S_a}{g}=1+15T\ (T<0.10\,\text{s}),\ \text{then as above (6.4.2}(b)\text{)}$$

$$\mathbf{F}_k=\mathbf{M}\boldsymbol{\phi}_k\Gamma_kA_kg,\qquad \mathbf{U}_k=\boldsymbol{\phi}_k\Gamma_kA_kg/\omega_k^2$$

All modes are combined; the number needed for $90\,\%$ of the mass is reported (7.7.5.2). Storey shears of every frame and of the building are combined by CQC (7.7.5.3, $\zeta=0.05$):

$$\lambda=\sqrt{\sum_i\sum_j\lambda_i\rho_{ij}\lambda_j},\qquad \rho_{ij}=\frac{8\zeta^2(1+\beta)\beta^{1.5}}{(1-\beta^2)^2+4\zeta^2\beta(1+\beta)^2},\ \beta=\omega_j/\omega_i;\qquad F_i=V_i-V_{i+1}.$$

Scaling (7.7.3): forces $\times\max(1,\ \bar V_B/V_B^{\text{dyn}})$ per direction, $\bar V_B=$ static base shear with T_a (including Table 7). Displacements are taken from the scaled forces (conservative; Amd 2 allows unscaled).

5.5 Torsion — design eccentricity (IS 1893 7.8.2)

e_s = centre of mass − centre of resistance; the centre of resistance at each floor is the stiffness-weighted position of the frames, with storey stiffness $k = V/\delta$ of each frame under the static force pattern.

$$e_d = 1.5e_s + 0.05b \quad \text{or} \quad e_d = e_s - 0.05b \quad \text{(more severe)}, \qquad b = \text{plan dimension} \perp \text{force}.$$

Static method: force at $x_{cm}/y_{cm} + \Delta$ with $\Delta = 0.5e_s + 0.05b\,s$ or $-0.05b\,s$, $s = \text{sign}(e_s)$. Response spectrum: e_s is in the dynamic model; the additional torsion $F_i\Delta$ is added. Each frame takes the more severe case; torsion loads the frames of the other direction too (column axial forces from both planes).

5.6 Drift (IS 1893 7.11.1.1, IS 456 20.5)

Per column of every frame under the unfactored design forces: $\delta/h \le 0.004$ (EQ, FAIL; all column stacks grow one size and the design repeats — D-16); wind top sway $\le H/500$ (WARN).

5.7 Irregularities (IS 1893 Tables 5, 6 with Amendment 2)

Check	Rule and result
Torsional (5(i))	$\Delta_{\max}/\Delta_{\text{ave}}$ of total displacements at the plan ends (static forces with e_d). > 1.4 FAIL (revise configuration); 1.2–1.4: FAIL unless the first torsional period is below the first X and Y periods, else WARN.
Re-entrant corner (5(ii))	projection $> 15\,\%$ (estimated $\sqrt{\text{bbox} - \text{footprint}}$): WARN — needs a flexible-diaphragm 3D analysis (not in StructOptima).
Openings (5(iii))	$> 50\,\%$: WARN, flexible diaphragm.
Soft storey (6(i))	storey stiffness $k_i = V_i/\delta_i$ more than $10\,\%$ below k_{i+1} (D-27): WARN if drift $\le 0.2\,\%$ in that storey and below (dynamic analysis performed), else FAIL.
Mass (6(ii))	$W_i > 1.5W_{i-1}$: handled by the dynamic analysis.
Floating column (6(vi))	FAIL (not permitted when part of the lateral system).
Irregular modes (6(vii))	first three modes $< 65\,\%$ mass in a direction: FAIL; Zones IV/V: $\lvert T_{1x} - T_{1y}\rvert < 0.1\,\max$: FAIL.
Method (7.7.1)	static method only for regular buildings < 15 m in Zone II; dynamic analysis is always performed.
Structural system (Table 9 Note 1)	OMRF in Zones III–V: FAIL.

6 Wind (IS 875 Part 3:2015)

$V_z = V_b k_1 k_2 k_3$, $V_b = 33, 39, 44, 47, 50, 55$ m/s (zones 1–6), $k_1 = k_3 = 1$, k_2 from Table 2 (linear interpolation; 10 m value below 10 m); $p_z = 0.6V_z^2$ N/m^2; $K_d = K_a = K_c = 1$ (conservative). Storey force $F = p_z(C_{pe,ww} + |C_{pe,lw}|)B\,h_{\text{trib}}$, $C_{pe,ww} = 0.8$, $C_{pe,lw} = -0.3/-0.4/-0.5/-0.6$ for $h/b \le 0.5/1/2/ > 2$; applied at the plan centre of the 3D model. $h/w \ge 6$: FAIL (Table 5 gives up to $+0.95/-1.85$). Overturning: stable if $0.9W\,B/2 > 1.5\sum Fz$.

z (m)	TC1	TC2	TC3	TC4
10	1.05	1.00	0.91	0.80
15	1.09	1.05	0.97	0.80
20	1.12	1.07	1.01	0.80
30	1.15	1.12	1.06	0.97
50	1.20	1.17	1.12	1.10
100	1.26	1.24	1.20	1.20
150	1.30	1.28	1.24	1.24
200	1.32	1.30	1.27	1.27

z (m)	TC1	TC2	TC3	TC4
250	1.34	1.32	1.29	1.28
300	1.35	1.34	1.31	1.30
350	1.35	1.35	1.32	1.31
400	1.35	1.35	1.34	1.32
450	1.35	1.35	1.35	1.33
500	1.35	1.35	1.35	1.34

7 Section design (IS 456:2000)

7.1 Flexure

$$\frac{x_{u,\max}}{d} = 0.53,\ 0.48,\ 0.46 \text{ (Fe250, 415, 500)}, \qquad M_{u,\lim} = 0.36 f_{ck} b d^2 k(1 - 0.42k),\ k = \frac{x_{u,\max}}{d}$$

$$p_t = 50\frac{f_{ck}}{f_y}\left(1 - \sqrt{1 - \frac{4.598R}{f_{ck}}}\right),\ R = \frac{M_u}{bd^2}; \qquad A_{s,\min} = \frac{0.85bd}{f_y};\ A_{s,\max} = 0.04bD.$$

Doubly reinforced ($M_u > M_{u,\lim}$):

$$A_{st1} = \frac{0.36 f_{ck} b\, x_{u,\max}}{0.87 f_y}, \qquad \varepsilon_{sc} = 0.0035\left(1 - \frac{d'}{x_{u,\max}}\right),$$

$$A_{sc} = \frac{M_u - M_{u,\lim}}{(f_{sc} - 0.446 f_{ck})(d - d')}, \qquad A_{st2} = \frac{A_{sc}(f_{sc} - 0.446 f_{ck})}{0.87 f_y}.$$

- Concrete: parabola–rectangle, peak $0.446 f_{ck}$, $\varepsilon_0 = 0.002$, $\varepsilon_{cu} = 0.0035$.
- Cold-worked steel (SP 16 1.4): stress 0.80, 0.85, 0.90, 0.95, 0.975, 1.00 f_{yd} at inelastic strain 0, 0.0001, 0.0003, 0.0007, 0.0010, 0.0020.

7.2 Shear (IS 456 40, 26.5.1.5, 26.5.1.6)

$\tau_v = V_u/(bd)$; $\tau_v > \tau_{c,\max}$ ⇒ FAIL; $s = \min\Big(\frac{0.87 f_y A_{sv} d}{(\tau_v - \tau_c) bd},\ \frac{0.87 f_y A_{sv}}{0.4b},\ 0.75d,\ 300\Big)$, $f_y \le 415$ for stirrups; $s < 75$ mm ⇒ FAIL.

p_t	M15	M20	M25	M30	M35	M40+
≤0.15	.28	.28	.29	.29	.29	.30
0.25	.35	.36	.36	.37	.37	.38
0.50	.46	.48	.49	.50	.50	.51
0.75	.54	.56	.57	.59	.59	.60
1.00	.60	.62	.64	.66	.67	.68
1.25	.64	.67	.70	.71	.73	.74
1.50	.68	.72	.74	.76	.78	.79
1.75	.71	.75	.78	.80	.82	.84
2.00	.71	.79	.82	.84	.86	.88
2.25	.71	.81	.85	.88	.90	.92
2.50	.71	.82	.88	.91	.93	.95
2.75	.71	.82	.90	.94	.96	.98
≥3.00	.71	.82	.92	.96	.99	1.01
$\tau_{c,\max}$	2.5	2.8	3.1	3.5	3.7	4.0

IS 456 Tables 19, 20 (N/mm^2), linear interpolation; grades between columns use the lower column.

7.3 Columns (IS 456 25, 39)

$$e_{\min} = \frac{l}{500} + \frac{D}{30} \geq 20 \text{ mm (one axis at a time)}, \qquad P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc},$$

$$\alpha_n = \begin{cases} 1 & P_u/P_{uz} \leq 0.2 \\ 1 + \dfrac{P_u/P_{uz} - 0.2}{0.6} & 0.2 < P_u/P_{uz} < 0.8 \\ 2 & P_u/P_{uz} \geq 0.8 \end{cases}$$

$$\left(\frac{M_{ux}}{M_{ux1}}\right)^{\alpha_n} + \left(\frac{M_{uy}}{M_{uy1}}\right)^{\alpha_n} \leq 1, \qquad M_a = \frac{P_u D}{2000}\left(\frac{l_e}{D}\right)^2 \text{ for } l_e/D > 12 \ (k = 1), \ l_e = 1.0\, l \text{ (D-04)}.$$

M_{u1} from strain-compatibility interaction curves (120 fibres; neutral axis outside the section: 0.002 at $3h/7$). Steel 0.8–4 % (IS 456 allows 6 %); ≥ 8 bars, $\phi \geq 16$; ties: pitch $\leq \min(b, D, 16\phi, 300)$, 150 mm at the ground storey (project rule); $\phi_{\text{tie}} \geq \max(\phi/4, 6)$.

Starting sizes: at least 300 × 450 (project rule), 400 × 600 for interior columns over 400 kN; 300 mm minimum in Zones III–V (IS 13920 7.1.1). Failing stacks grow one preferred size (230, 300, 350, 380, 400, 450, 500, …).

Columns matching 230 mm walls (D-29, residential, on by default, Setup → Materials):

- only in Zone II with OMRF: IS 13920:2016 1.1.1 makes it mandatory in Zones III–V and 7.1.1 requires $\geq \max(300,\ 20d_b)$; SMRF relies on IS 13920 detailing;
- only columns standing in a full-brick wall line (wall $\geq$ 200 mm, centre line within 120 mm of the column centre): side across the wall = 230 mm, side along the wall L from $1.5P \leq 0.4 f_{ck}(0.992A_g) + 0.67 f_y(0.008A_g)$, rounded up to a preferred size, $450 \leq L \leq 900$ mm (IS 456 32.1: longer than 4×230 is a wall);
- where walls meet, the long side goes to the direction with fewer long sides so far (balanced stiffness);
- growth: along the wall up to 900 mm; when bending about the 230 side governs ($M_{uy}/M_{uy1} > M_{ux}/M_{ux1}$ for a wall along x) or beyond 900, both sides grow and the column is reported "Column wider than wall";
- beams framing into a 230 side are made 230 wide (depth $\leq$ 750); a deeper (300 wide) beam widens the column;
- fire: IS 456 Fig. 1 *50 % exposed* with the dimension along the wall (2 h: 200 mm); free-standing columns *fully exposed* (2 h: 300 mm).

Beam wider than column (all designs): WARN when a beam is wider than the column side across it.

8 Members

- **Beams:** one design per plan beam for the worst floor (every floor shows the same values); $d = D - c - 8 - \phi/2$ (c from exposure/fire); faces designed for sagging/hogging envelopes; FAIL if > 4 % per face or > 2 layers; depth grows by 50 mm (from $\max(L/12, 300)$) while FAIL or $\rho > 2.5$ %, up to 1200 mm; width 300 when $D > 750$; WARN if $L/d > 20$ (simply supported) / 26 (continuous). DXF beams start at 300×600 (no size in plan drawings).
- **Slabs:** $t = \max\left(\lceil (l_x/26 + c')/5 \rceil 5,\ 125,\ t_{\text{occ}}\right)$ with $c' = \max(30, c_{\text{slab}} + 10)$; two-way: $M = \alpha w l_x^2$ with IS 456 Table 26 (case by discontinuous short / long edges, $l_y/l_x \in [1, 2]$); one-way ($l_y/l_x > 2$): $wl^2/8$; $w = 1.5(25t + SDL + LL)$; steel ≥ 0.12 %, spacing $\leq \min(3d, 300)$.
- **Staircase:** dog-legged flight, waist $\geq \max(\text{span}/25, 150)$, $w = 25t\sqrt{R^2 + T^2}/T + 25R/2 + LL + FF$, $M_u = 1.5wL^2/8$.
- **Footings:** side $= \sqrt{1.1P/\text{SBC}} \geq 1.5$ m ($P \leq 400$) / 2.0 m; FAIL > 10 m; thickness from 300/350 mm in 50 mm steps until punching at $d/2$ ($\tau \leq k_s 0.25\sqrt{f_{ck}}$, $k_s = 0.5 + \beta_c \leq 1$), bending at the column face and one-way shear at d pass; $d = t - c - 12$, $c \geq 50$; steel ≥ 0.12 %; WARN if $e = M/P > B/6$ or edge pressure > SBC (sizing uses axial load).

Case	Panel	1.0	1.1	1.2	1.3	1.4	1.5	1.75	2.0	α_y^-	α_y^+
1	Interior − / +	.032	.037	.043	.047	.051	.053	.060	.065	.032	.024
		.024	.028	.032	.036	.039	.041	.045	.049		
2	One short edge discont.	.037	.043	.048	.051	.055	.057	.064	.068	.037	.028
		.028	.032	.036	.039	.041	.044	.048	.052		
3	One long edge discont.	.037	.044	.052	.057	.063	.067	.077	.085	.037	.028
		.028	.033	.039	.044	.047	.051	.059	.065		
4	Two adjacent edges	.047	.053	.060	.065	.071	.075	.084	.091	.047	.035
		.035	.040	.045	.049	.053	.056	.063	.069		

Case	Panel	1.0	1.1	1.2	1.3	1.4	1.5	1.75	2.0	α_y^-	α_y^+
5	Two short edges	.045	.049	.052	.056	.059	.060	.065	.069	–	.035
		.035	.037	.040	.043	.044	.045	.049	.052		
6	Two long edges (+ only)	.035	.043	.051	.057	.063	.068	.080	.088	.045	.035
7	Three edges, long cont.	.057	.064	.071	.076	.080	.084	.091	.097	–	.043
		.043	.048	.053	.057	.060	.064	.069	.073		
8	Three edges, short cont. (+)	.043	.051	.059	.065	.071	.076	.087	.096	.057	.043
9	Four edges (+)	.056	.064	.072	.079	.085	.089	.100	.107	–	.056

IS 456 Table 26, short-span coefficients by l_y/l_x (first row negative, second positive) and long-span coefficients. Case 4 $\alpha_y^- = 0.047$ (corrected 20 Oct 2026, was 0.045).

9 Ductile design checks (IS 13920:2016, Amendments 1 and 2)

Applied in Zones III–V and wherever SMRF is selected.

Clause	Rule
5.2	minimum M20; M25 for buildings > 15 m in Zones III–V (FAIL)
5.3.1 (Amd 2)	elongation $\geq 14.5\,\%$, $1.15 \leq TS/YS \leq 1.25$, grades 415/500/550 (WARN: use D-grade bars, mill certificates; plain Fe 500 has 12 %)
7.1 (Amd 2)	factored axial stress under EQ combinations $\leq 0.40 f_{ck}$ (FAIL)
7.1.1, 7.1.2 (Amd 1)	column least dimension $\geq \max(300,\ 20\phi_{\text{beam,max}})$; $b/D \geq 0.4$ (FAIL)
7.2.1, 7.2.1.3	$\sum M_c \geq 1.4 \sum M_b$; roof joints checked up to 4 storeys; FAIL for SMRF, WARN for OMRF
9.1.1, 9.1.2 (Amd 2)	$\tau_{jc} = k\sqrt{f_{ck}}$, $k = 1.5/1.2/1.0$ for beams on 4/3/other faces; $A_{ej} = b_j h_c$, $b_j = \min(b_b,\ b_c + 0.5h_c)$ if $b_c < b_b$, else $\min(b_c,\ b_b + 0.5h_c)$ (second form not printed in full — UNVERIFIED, conservative); demand $1.25 f_y (A_{st,top} + A_{st,bot})$
6.3.5 (Amd 1)	beam end zone $2d$: $s \leq \min(d/4,\ 6\phi,\ 100)$
7.6.1	column $l_o = \max(D,\ l/6,\ 450)$, $s \leq \min(b/4,\ 6\phi,\ 100)$ (stricter than Amd 1's 6ϕ)
3.5, 6.3.1 (Amd 1, 2)	135° hook extension $8\phi \geq 75$ mm
7.6.1(c)	A_{sh} — **not designed**

10 Durability and fire

	Mild	Moderate	Severe	Very severe	Extreme
Cover, Table 16 (mm)	20	30	45	50	75
Minimum grade, Table 5	M20	M25	M30	M35	M40
Max free w/c	0.55	0.50	0.45	0.45	0.40
Min cement (kg/m^3)	300	300	320	340	360

Severe / very severe with M35+: −5 mm (Table 16 Note 3).

NBC Table 1 (min)	Type 1	Type 2	Type 3	Type 4
Columns supporting > 1 floor	240	120	120	120
One floor / roof only	180	90	60	60
Floors	120	90	60	60

Hours	Beam b	Slab D	Column (full)	Beam cover SS/cont	Slab cover SS/cont	Column cover
0.5	200	75	150	20/20	20/20	40
1	200	95	200	20/20	20/20	40
1.5	200	110	250	20/20	25/20	40
2	200	125	300	40/30	35/25	40
3	240	150	400	60/40	45/35	40
4	280	170	450	70/50	55/45	40

IS 456 Fig. 1 and Table 16A (mm). Covers used: beams max(Table 16, Table 16A continuous); slabs max(Table 16, Table 16A simply supported); columns max(Table 16, 40); footings max(Table 16, 50). Fire or slenderness failures are FAIL (column $l/D \le 60$; beam $L/b \le \min(60, 250b/D)$).

11 DXF import rules and edge cases

11.1 What each layer holds (since 20 Oct 2026, D-28)

Every layer gets a role — Columns, Beams, Walls, *Decide by shape* or Ignore — suggested from two kinds of evidence and shown on the Setup page with the reason; the engineer can change any role, and a confirmed mapping is used exactly as given (layers not listed are ignored).

1. **Name** (whole words of the layer name): columns COL, COLS, COLM, COLMN, CLMN, COLUMN(S), PILLAR(S), STANCHION(S), POST(S); beams BEAM(S), BM, GIRDER(S), JOIST(S); walls WALL(S) (also inside a word), MASONRY, BRICK, BLOCKWORK, PARTITION(S). Ignored: annotation (TEXT, TAG, DIM, HATCH, IDEN, NOTE, TITLE, …), grid / centre lines (GRID, AXIS, CENTRE, CL), doors, windows, furniture, services, stairs. **Plinth and lintel beams** (PLINTH, LINTEL, PB, LB, SUNSHADE, CHAJJA) are ignored by default: they are not floor beams of the frame.
2. **What is drawn** (in mm after the units, u m per drawing unit):
   - column-sized outline: closed polyline or circle with $150 \le \min(a,b)$ and $\max(a,b) \le 1500$ mm;
   - beam-like line: both ends within $\frac{1}{2}\max(a,b) + 150$ mm of two different column centres and both centres within $\max(\frac{1}{4}\min(a,b), 50\text{ mm})$ of the line;
   - grid-like line: passes over column centres (same tolerance) but at least one end is away from a column;
   - wall pair: two lines within 2° of parallel, 75–500 mm apart, overlapping ≥ 100 mm (≥ 500 mm on *decide by shape* layers, so tag leaders and ticks are not walls).

Suggestion when the name says nothing: ≥ 2 rectangular column outlines and no other structure ⇒ Columns; outlines and lines together ⇒ Decide by shape; $\ge 60\,\%$ of the lines beam-like ⇒ Beams; $\ge 50\,\%$ of the remaining lines in wall pairs ⇒ Walls; mostly grid-like or text only ⇒ Ignore. Circles alone on an unnamed layer are taken as symbols (grid bubbles). If no layer is walls or columns, lines of unrecognised layers are taken as walls (warned). *Decide by shape* layers: column outlines → columns, beam-like lines → beams, wall pairs → walls, the rest ignored.

Confidence shown: high (name and drawing agree), medium (drawing only), low (name without matching geometry, or a guess).

11.2 Geometry rules

- Blocks (INSERT) are exploded recursively (depth 8); entities on layer 0 inside a block take the INSERT's layer; unbound xrefs and missing blocks are reported.

- Columns: sides from the frame of the longest edge (rotated blocks); circle ⇒ square of equal area (0.886× diameter, WARN); duplicates within $\frac{1}{4}$ of the smaller side counted once (WARN); outlines with a side < 150 mm or > 3 m on column layers are not columns (WARN).
- Units from $INSUNITS (in, ft, mm, cm, m). **Unitless or implausible** (< 1 m or > 1000 m extent; every R12 file) **(D-35, 21 Oct 2026):** the first of mm, m, cm, inch, ft for which (a) the extent is 2–1000 m, (b) the median distance from each column-sized outline / circle (150 mm – 1.5 m) to the nearest distinct one is 1.5–15 m, with (c) the most column-sized outlines; without such evidence millimetres if coordinates exceed 500, else metres. The unit and the reason are a WARN. (Coordinate size alone read inch / foot / cm drawings, and metre drawings with survey coordinates, at the wrong scale, N55.)
- **Rotated plans (D-34):** when the rectangular column outlines (75 % within 0.5° of their median), else the beam / wall lines (length-weighted resultant of $4\theta \geq 0.95$), agree on an angle θ to the axes with $|\theta| \geq 0.05°$ (modulo 90°), the drawing is rotated by $-\theta$ before reading (WARN).
- Entities with extrusion $(0, 0, -1)$ (mirrored blocks) are rewritten upright before reading (N53).
- **Circular columns on layers without a column name (D-36):** a column-sized circle is a column when two or more line ends lie within $\frac{1}{2}d + 150$ mm of its centre; a lone circle (grid bubble) is not.
- Coordinates moved so the first grid intersection is $(0, 0)$ (WARN when moved by > 1 m); column centres within 150 mm merge into one grid line (WARN with the misalignment).
- Beams: LINE and every polyline segment; ends within half a column +150 mm of a column centre are moved to it; duplicate and zero-length beams dropped; loose ends reported.
- Wall thickness: most frequent perpendicular gap 75–500 mm between parallel overlapping lines (≥ 0.1 m), unit-aware; 230 mm if none.
- **Geometry bound (D-32, 21 Oct 2026):** before any framing, every column / beam / wall coordinate must be finite and the raw geometry must span at most 10^6 drawing units (1000 m even in millimetres, the most permissive unit); after the units, the structure must span at most $L_{\max} = 1000$ m in X and in Y (the same bound that decides whether $INSUNITS is plausible). Otherwise a clear input error naming the extent; nothing is designed. Only the extent is bounded: survey coordinates far from the origin are accepted. (A line 10^{11} units long made the automatic column placement create 3×10^8 nodes, N50.)
- No column found ⇒ automatic column placement from walls; nothing usable ⇒ clear input error.
- **Drawing vs design:** columns drawn at a size other than the designed one give a WARN listing drawn and designed sizes (e.g. 230×230 drawn, 300×450 designed).
- **Room labels outside the structure:** a room-name text inside the column extent but in no slab panel is reported (the area may be missing its columns / beams).

12 Other edge cases and fallbacks

Situation	Behaviour
No SBC entered	run blocked (no default)
Site parameter changed after confirming	confirmation cleared
Inputs changed after a run	every result page says the results are out of date
Frame cannot be solved	listed (FAIL); its beams use $wL^2/8$ (WARN)
Beam not on a grid line	simply supported fallback (WARN)
Column on a frame line in one direction only	no frame moments in the other (minimum eccentricity, WARN)
Design not converged in 6 iterations	last results shown with a WARN
Zero-load column	nominal footing with a warning
Tension at a column base	footing for $\lvert P\rvert$, uplift not checked (WARN)
Footings overlapping	WARN (combined footing needed)
Column drawn smaller / larger than designed	WARN “Drawing vs design”
Room label outside the structure read	note on the Setup page
Layer role changed by the engineer	results marked out of date
Beam deflection without analysis load	not evaluated (no assumed load)

Situation	Behaviour
Floors without torsional restraint	translation-only sharing (WARN)
Non-finite coordinate, or structure wider than 1000 m	input error before any calculation (D-32)
Columns all on one line, or no slab panel enclosed by the drawn columns / beams / walls	input error (D-33)
Floor without rotational mass in the 3D model	translation-only sharing (WARN), no torsion
Non-finite design force (NaN / ∞) in a column or beam check	design stops with a numerical error (`FloatingPointError`); never a pass, never an input error (N51, N52)
Non-finite W or V_B	FAIL "Numerical failure"
P_u outside the interaction diagram ($P_u > P_{uz}$)	ratio ∞, FAIL; message gives P_u and P_{uz}; every output shows "beyond capacity" (D-38)
Gross stiffness selected	WARN (6.4.3.1)
BBS	one typical floor and ground-storey columns; default bars when a description cannot be parsed (KD-21)

13 Not performed (must be checked separately)

Flexible diaphragms and diaphragm / collector / chord design; time history, site-specific spectra, soil–structure interaction, P–Δ; $100 + 30$ % and vertical earthquake; pattern live loading; effective length (Annex E); IS 13920 confining steel, capacity-design shear, joint reinforcement; development length, laps and curtailment in design; beam torsion and deflection beyond basic L/d; footing sizing for base moments and settlement; corridor / stair / balcony loads; wind k_3, k_4.

14 Verification evidence

Benchmark	Result
SP 16 Tables C, D ($M_{u,\lim}$)	within 0.8 % (IS 456 rounds $x_{u,\max}/d$)
SP 16 Tables A, F (steel stress)	within 0.5 % / 1.2 %
SP 16 Examples 1, 2, 3 (beams, slab)	10.02 cm^2; 3.8 cm^2/m; 18.85 / 7.05 cm^2 reproduced
SP 16 Example 6 (column)	capacity within 10 %
SP 16 Example 9 (slender column)	P_{uz}, M_a, $e_{\min}$ within 1 %
SP 16 Example 10 (shear)	$\tau_v = 1.07$, $\tau_c = 0.55$, 140 mm
3D model, symmetric, $\pm 0.05b$	worse frame $0.525F$; $\Delta_{\max}/\Delta_{\text{ave}} = 1.05$
3D model, eccentric stiffness	50.5 kN on the flexible frame (hand)
Two-storey shear building	$\omega^2 = (k/m)(3 \mp \sqrt{5})/2$
Portal drift	3.356 mm vs 3.333 mm by hand
Worked hand check, `complex_house.dxf`	every value of Appendix A reproduced (`test_hand_check_complex_house.py`)
Test suite	809 passed, 3 skipped (20 Oct 2026); CI green on Python 3.11–3.13 (at 774)

A Worked hand check: `samples/plans/complex_house.dxf`

Purpose. Check the site against a calculator on one non-trivial building. The same numbers are locked in `tests/test_hand_check_complex_house.py`; if the site changes, that test fails.

How to reproduce on the site. Setup page → *CAD drawing (DXF)* → upload `samples/plans/complex_house.dxf`; auto-framing off; 3 storeys of 3.0 m; Residential; SBC 150; Zone II; soil II; SMRF; $I = 1.0$; M25; Fe500; exposure

Moderate; NBC Type 2; wind zone 3, terrain 2; cracked sections and fixed bases on. Run. Each value below is on the page named in brackets.

A.1 Read the drawing (Members, Drawings & 3D)

L-shaped plan 12 m × 8 m with the 4 m × 4 m corner at $x = 8$–12, $y = 4$–8 missing; grid 4 m both ways; 11 columns, 15 beams (all 300 × 600 start size). Check: every column of the DXF is shown, nothing in the missing corner. *Found during this check (20 Oct): the drawing's columns are* 230×230 *but the design starts from* 300×450 *without saying so; the label "Master Bed" lies outside the building. Both are now reported (Section 11).* With SMRF the 230 mm wall columns (D-29) do not apply.

A.2 Slab panel S1 (Members → Slabs) — Sections 4.1, 8, Table 26

$$
\begin{aligned}
c_{\text{slab}} &= \max(\text{Table 16 Moderate } 30,\ \text{Table 16A 1.5 h SS } 25) = 30 \text{ mm} \\
c' &= \max(30,\ c_{\text{slab}} + 10) = 40 \text{ mm} \\
t &= \lceil (4000/26 + 40)/5 \rceil \cdot 5 = \mathbf{195} \text{ mm} \\
w &= 1.5\,(25 \times 0.195 + 1.0 + 1.5 + 0.5 + 2.0) = \mathbf{14.81} \text{ kN/m}^2 \\
\text{Case 4 (corner), } l_y/l_x = 1: \quad M_x^- &= 0.047 \times 14.81 \times 4^2 = \mathbf{11.14}, \quad M_x^+ = 0.035 \times 14.81 \times 4^2 = \mathbf{8.30} \text{ kN m/m}
\end{aligned}
$$

A.3 Seismic weight and base shear (Checks → Seismic) — Sections 4.2, 5.3

Per storey: slab area 80 m^2; beams 60 m; walls 60 m; columns $9 \times (0.30 \times 0.45) + 2 \times (0.40 \times 0.60)$.

$$
\begin{aligned}
\text{wall} &= 0.23 \times (3.0 - 0.125 - 0.400) \times (1 - 0.33) \times 19.2 = 7.323 \text{ kN/m} \\
\text{web} &= 0.3 \times (0.6 - 0.125) \times 25 = 3.5625 \text{ kN/m} \\
\text{overlap} &= 20 \text{ panel edges} \times 0.15 \times 1.75 \times 4 = 21.0 \text{ kN (N46)} \\
W_{1,2} &= 80(3.125 + 3.0 + 1.75) + 0.25 \times 2.0 \times 80 + 60(3.5625) - 21.0 + 60(7.323) + 127.1 = \mathbf{1429.2} \text{ kN} \\
W_3 \text{ (roof)} &= 80(3.125 + 1.5 + 1.75) + 60(3.5625) - 21.0 + (0.5 + 0.33)\,60(7.323) + 63.6 = \mathbf{1131.0} \text{ kN} \\
W &= \mathbf{3989.5} \text{ kN}, \quad T_a = \min(0.075 \times 9^{0.75},\ 0.09 \times 9/\sqrt{12}) = \mathbf{0.234} \text{ s} \\
S_a/g &= 2.5, \quad A_h = \tfrac{0.10}{2} \cdot \tfrac{1.0}{5} \cdot 2.5 = 0.025, \quad V_B = 0.025 \times 3989.5 = \mathbf{99.7} \text{ kN} \\
Q_3 &= 99.7 \times \frac{1131.0 \times 9^2}{1429.2(3^2 + 6^2) + 1131.0 \times 9^2} = \mathbf{58.6} \text{ kN}
\end{aligned}
$$

Dynamic (cannot be done by hand): check only that *scale factor* $= V_B/V_B^{\text{dyn}}$ (99.7/80.0 = 1.247 in X) and that the design V_B equals the static 99.7 kN (IS 1893 7.7.3). Note: $1.75 = 25 \times (0.195 - 0.125)$ is the thicker designed slab; it runs to the beam centre lines, so the half beam width per panel edge is taken off the webs (N46, fixed 20 Oct; before the fix $W_{1,2} = 1450.2$, $V_B = 101.3$).

A.4 Beam B5, interior, $y = 4$, $x = 4$–8 (Members → Beams) — Sections 4.1, 7.1, 7.2

$$
w_{\text{service}} = 2 \times \tfrac{4}{3} \times (3.125 + 3.0 + 1.75 + 2.0) + (3.5625 - 2 \times 0.15 \times 1.75) + 7.323 = \mathbf{36.69} \text{ kN/m}
$$

The moments come from the frame analysis (not by hand; an independent 2D frame program such as Ftool can be used: 3-storey frame on $y = 4$, spans 4/4/4 m). The site designs each plan beam for its worst floor, so every floor shows the same values: $M_u^- = 97.84$ kN m, $V_u = 112.04$ kN. Check the section design from these:

$$
\begin{aligned}
d &= 600 - 30 - 8 - 12/2 = 556 \text{ mm}, \quad R = \frac{97.84 \times 10^6}{300 \times 556^2} = 1.055 \\
p_t &= 50 \tfrac{25}{500} \left(1 - \sqrt{1 - 4.598 \times 1.055/25}\right) = 0.256\,\% \ \Rightarrow\ A_{st} = \mathbf{426} \text{ mm}^2 \text{ (4-12 = 452 provided)} \\
\tau_v &= \frac{112.04 \times 10^3}{300 \times 556} = \mathbf{0.672}, \quad p_t = \frac{100 \times 339}{300 \times 556} = 0.203\,\% \Rightarrow \tau_c = 0.29 + 0.53 \times 0.07 = \mathbf{0.327} \text{ N/mm}^2 \\
s &= \min(351,\ 302,\ 417,\ 300) = \mathbf{300} \text{ mm}; \quad \text{end zone } \min(556/4,\ 6 \times 12,\ 100) = 72 \to \mathbf{70} \text{ mm}
\end{aligned}
$$

A.5 Column C5, ground storey (Members → Columns, math log) — Section 7.3

400 × 600, 10-16 ($A_s = 2011$ mm^2, 0.84 %). Gravity: $P_u = 1.5 \times 705.4 = \mathbf{1058}$ kN. $P_{uz} = 0.45 \times 25 \times (240000 - 2011) + 0.75 \times 500 \times 2011 = \mathbf{3431}$ kN. Governing $0.9DL + 1.5EQx$: $P_u = 551.0$ kN, $M_x = 50.8$ kN m (depth 400 mm in X). SP 16 **Chart 49** (Fe500, four sides, $d'/D = 56/400 \approx 0.15$): $P_u/(f_{ck}bD) = 551.0 \times 10^3/(25 \times 600 \times 400) = 0.092$, $p/f_{ck} = 0.84/25 = 0.034 \Rightarrow$ read $M_u/(f_{ck}bD^2) \approx 0.08$, i.e. $M_{ux1} \approx 0.08 \times 25 \times 600 \times 400^2 = \mathbf{190}$ kN m (site 199 with its actual bar layout: 198.3). Ratio $50.8/198.3 = 0.26$ plus the y term $\Rightarrow 0.31 \leq 1$: PASS.

A.6 Footing under C5 (Members → Footings) — Section 8

$$P = 705.4 + 58.6 \text{ (plinth wall)} = 764.0 \text{ kN}, \quad B = \sqrt{1.1 \times 764.0/150} = 2.37 \rightarrow \mathbf{2.40} \text{ m}$$

$$d = 500 - 50 - 12 = 438, \quad q_u = 1146.0/2.40^2 = 199.0 \text{ kN/m}^2$$

$$V_u = 1146.0 - 199.0 \times 0.838 \times 1.038 = 973 \text{ kN}, \quad \tau_v = \frac{973 \times 10^3}{2(838 + 1038) \times 438} = 0.59 \leq 1.0 \times 0.25\sqrt{25} = 1.25 \text{ OK}$$

$$M_u = 199.0 \times 1.0^2/2 = 99.5 \text{ kN m/m} \Rightarrow A_{st} = 535 < 0.12\,\% = \mathbf{600} \text{ mm}^2\text{/m (12@180)}$$

A.7 Strong column–weak beam, roof of C1 in X (Checks → Joints) — Section 9

Beam B1, 3-12 top (339 mm^2), $d = 600 - 30 - 16 = 554$: $x_u = \frac{0.87 \times 500 \times 339}{0.36 \times 25 \times 300} = 54.7$, $M_b = 0.87 \times 500 \times 339 \times (554 - 0.42 \times 54.7) = \mathbf{78.4}$ kN m. Column C1 at the roof: one column only, 300 × 450 with 300 in X, 8-16, $P_u = 48$ kN: SP 16 Chart 50 ($d'/D = 56/300 \approx 0.20$), $P_u/(f_{ck}bD) = 0.014$, $p/f_{ck} = 0.048 \Rightarrow M_c \approx 76.5$ kN m. $\sum M_c / \sum M_b = 76.5/78.4 = \mathbf{0.98} < 1.4$: **FAIL** — correct: IS 13920 7.2.1.3 waives roof joints only above 4 storeys. 24 of 66 joints fail in this building; the site reports them but does not enlarge the columns (rotate C1 so 450 is along X, enlarge the columns, or detail them as gravity columns per 7.2.1).

A.8 What you cannot check by hand

Frame moments (use an independent frame program), modal periods and torsion ratio (3D model; compare with ETABS / STAAD), the L-shape flexible-diaphragm check (WARN on the site).